

ALBERT JOHN C. DIAZ

(+63) 9199277304 | diaz.albertjohn123@gmail.com | Philippines | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience in embedded systems, IoT firmware development, full-stack software engineering, and networking. Proven ability to design and integrate hardware-software solutions, utilizing microcontrollers, IoT protocols, and machine learning to build end-to-end monitoring and recommendation systems. Proficient in Python, C/C++, JavaScript (React, Node.js), and PostgreSQL, with a solid foundation in network fundamentals (CCNA). Civil Service Professional-eligible, recognized for strong technical documentation, cross-functional team leadership, and a highly adaptable, detail-oriented approach to complex problem-solving. Open to relocation and nationwide opportunities.

EDUCATION

Bachelor of Science in Computer Engineering

Graduated: June 2026

University of Science and Technology of Southern Philippines (USTP)

Dean's Lister (2022–2023)

WORK EXPERIENCE

Full-Stack & IT Support Intern

January 2026 – March 2026

Columbia Computer Center Inc. – CDO Branch

- Developed and maintained full-stack web applications using React, TypeScript, Node.js, and PostgreSQL, applying iterative delivery practices aligned with Agile workflows.
- Performed hardware diagnostics and computer repair tasks, gaining hands-on exposure to digital circuit troubleshooting and system integration.
- Contributed to job order management processes, coordinating between technical and administrative teams to ensure timely service delivery.
- Documented technical procedures and system configurations to support operational compliance and knowledge transfer.

PROJECTS

TANIM: AN AGRICULTURAL SYSTEM WITH INTEGRATED IOT SOIL HEALTH MONITORING DEVICE AND CROP RECOMMENDATION USING LIGHT GBM

May 2025 – April 2026

ESP32 · Modbus-RTU · RS485/UART · LoRa · GPS · Light GBM · Embedded C · Solar Power

- Led hardware and firmware development as Hardware Lead; designed and implemented embedded firmware for ESP32 microcontrollers to interface with a 7-in-1 soil sensor (pH, NPK, moisture, temperature, salinity) over Modbus-RTU via RS485/UART, handling register-level read commands, response parsing, and fault detection for robust real-time soil data acquisition across 3 field nodes.
- Architected a long-range wireless communication layer using LoRa with a TDMA scheme and stop-and-wait ACK protocol (1.5-second timeout window), enabling reliable IoT data pipelines across a 3-node, 1-hectare agricultural deployment with limited cellular coverage.
- Integrated a GPS module for geo-tagged soil readings, enabling per-location tracking of soil health data across multiple farm plots in Bukidnon and powering a GIS-based zoning dashboard for Department of Agriculture officials.
- Implemented solar power management firmware using three 5V 1A solar panels charging four Lithium-Ion 18650 (2500mAh) batteries, regulating sensor polling intervals under low-charge conditions to ensure continuous field operation without manual intervention.

Job Order Management System

February 2026 – March 2026

React · TypeScript · Node.js · Express.js · PostgreSQL · Tailwind CSS · Git

- Built a production-grade web application with structured state management, role-based access, and real-time status monitoring, demonstrating systematic software development practices.
- Implemented end-to-end testing and validation procedures, ensuring system reliability and traceability of repair and service workflows.

- Maintained version control using Git with documented commit history, supporting collaborative development and code review processes.

IoT-Based Forest Monitoring System

March 2025 – June 2025

ESP32 · IoT · Embedded Firmware · Sound Sensor · ESP32-CAM · HTTP/Wi-Fi

- Designed and implemented event-driven, state-machine firmware (IDLE → SENSING → DETECTION → ALERT → RESET) on ESP32 microcontrollers to continuously poll a KY-038 audio sensor and classify environmental sounds in real time, achieving >90% detection accuracy in quiet environments and >80% in noisy conditions.
- Architected threshold-based alert pipeline with 100–500ms average response time, triggering buzzer/LED alerts and coordinating ESP32-CAM image capture on detection events, then uploading event data to Firebase with a >95% transmission success rate on stable networks.
- Integrated ESP32-CAM for visual verification of detected activities, coordinating data flow between peripheral modules via GPIO; system achieved an SUS score of 81 ("Excellent") in user acceptance testing with 5 respondents.

MART: Market of ART

March 2024 - December 2024

Kotlin · Android SDK · Mobile Application

- Led a 4-person team as project lead in designing and developing a dual-role e-commerce Android application in Kotlin connecting Artists/Artisans with customers, featuring auction bidding, a product feed, ratings and reviews, in-app messaging, and profile management.
- Implemented 14+ distinct activity screens with user authentication, role-based navigation, and session management; achieved 100% functional test pass rate and an SUS score of 84.5 ("Good") in usability testing.

CERTIFICATIONS & TRAINING

- Civil Service Eligibility – Professional Level, Civil Service Commission, Philippines (Passed: May 2026)
- Build a free website with WordPress - Coursera
- CISCO CCNA: Enterprise Networking, Security, and Automation
- CISCO CCNA: Switching, Routing, and Wireless Essentials
- CISCO: Introduction to Cybersecurity
- Educational Exposure: Profood International Corp., National Grid Corporation of the Philippines, Globe, OnSemi

TECHNICAL SKILLS

Embedded & Hardware: ESP32, Arduino, ARM-based microcontrollers, IoT sensors, I2C / SPI / UART protocols, Modbus-RTU, LoRa, TDMA, GPS, electronic diagnostics, PCB design, AutoCAD, Autodesk Eagle, EasyEDA, KiCAD

Networking: CCNA, Cisco IOS, TCP/IP, IPv4/IPv6 Subnetting, LAN/WAN, Modbus RTU, Network Troubleshooting,

Languages & Frameworks: C / C++ / Embedded C, Python, JavaScript / TypeScript, React, Node.js, Express.js, REST APIs, Kotlin, Java, HTML, CSS, Tailwind CSS, MATLAB, Verilog / VHDL, Assembly (MIPS), LaTeX

Databases & DevOps: PostgreSQL, MySQL, Firebase, Supabase, Git / GitHub, Docker, Linux, VSCode, Jupyter Notebook, Google Workspace, Microsoft Office

Soft Skills: academic writing, adaptable, attention to detail, communication, cross-functional collaboration, fast learner, presentation, problem-solving, team leadership, research, technical documentation

LEADERSHIP & ACTIVITIES

- Year Representative, Institute of Computer Engineers of the Philippines Student Edition – USTP (2024–Present)
- Mobile Legends Representative (Captain), ICePEP.SE – USTP Regionals & National Events (2023–2025)